

CANDIDATE CORRECTIONS TO MANUSCRIPT

- [Page 1]:
 - Across Eumetazoa, oxygen-sensing and hypoxia-responsive regulation **are** coordinated by the conserved hypoxia-inducible factor (HIF) pathway.
 - **Anthropogenic climatic changes are contributing to increases in the incidence of severe hypoxia in aquatic ecosystems.**
 - In this study, we show that the phylum Platyhelminthes has likely lost core components of the HIF pathway, yet still survives hypoxic stress and **mounts robust gene regulatory responses.**
 - **Nonetheless**, we find that planarians are tolerant to severe hypoxia for at least 24 hours.
 - An oxygen-dependent degradation (ODD) domain containing a proline residue within an 'AP' or 'SP' motif across phyla recognised by the HIF prolyl hydroxylase EGLN.
 - [Page 3]:
 - Causing decreases in **metabolic rate**, locomotion, survival, growth or reproduction (Han & Hagiwara, 2023; Lee *et al.*, 2023; Vaquer-Sunyer *et al.*, 2008).
 - This led us to examine the gene regulatory response to hypoxia to begin to understand how **low oxygen stress** is managed in the absence of the canonical HIF pathway.
 - [Page 4]:
 - Figure 1: Phylogeny of metazoan species from which **HIF pathway gene** anchor sequences etc.
 - Protein sequences from 25 species across 13 animal phyla were used to construct **p-HMMs for HIF- α , EGLN and VHL ortholog mining** across the proteomes of 9 platyhelminth species.
 - **Priapulida clades with Ecdysozoa**, not Lophotrochozoa.
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 - Figure 2: **Taenia crassiceps** was also sampled at the proteome level in Cestoda.
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 - **For HIF- α ortholog mining**, alignment using MAFFT iterative local-pair refinement was selected to maximise positional homology etc.
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 - Subsequent maximum-likelihood phylogenetic inference placed the *Clonorchis* sequence **in a phylogenetic tree of high-confidence metazoan VHL sequences** as an isolated branch outside other metazoan VHL clades etc.
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 - Figure 11A: **Amended**
- The figure displays a phylogenetic tree and a corresponding heatmap. The tree is rooted at the top and branches out to show relationships between different time points (0h, 24h, 36h, 12h) and replicates (R1, R2, R3) for three conditions: Time, Hypoxia, and Recovery. The heatmap below the tree uses a color scale from 0 to 800 to represent expression levels. The 'Time' condition shows a clear separation between the 0h and 24h/36h/12h groups. The 'Hypoxia' condition shows a more complex pattern of expression, with some branches showing higher expression than others. The 'Recovery' condition shows a pattern of expression that is distinct from the other two conditions.
- Figure 11C: Heatmaps **beside** the module assignments etc.
 - Figure 11D: Temporal trajectories of module expression showing mean FPKM **value** of genes etc.
- [Page 31]: These indirect sources of evidence are critical in a non-model metazoan with no transgenic technologies available, precluding the use of **CRE-reporter** constructs for directly testing the function and mapping the target genes of CREs.
- [Page 32]: Figure 18: Base-pair resolution view of hypoxia-specific Klf10/11 footprint in **h1SMcG0021003**.